

Week 10 · Working with an image or document

OPEN / WEEK 10

The €54 invoice

An invoice image lists two items at €20 and €5 tax, yet displays €54 total. A model reads €54 correctly. Should the payment pass?

INVOICE / INV-04

Quantity	2
Unit price	€20
Tax	€5
Listed total	€54

Calculated total = €45. Flag the mismatch.

First vote: what would you do?

QUESTION / WEEK 10

An invoice image lists two items at €20 and €5 tax, yet displays €54 total. A model reads €54 correctly. Should the payment pass?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 10

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

Multimodal input

2

Extraction

3

Verification

4

Uncertainty

1. Predict · Multimodal input

QUESTION / WEEK 10

What capability is represented by the supplied transcription?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A An upstream reading step, whose accuracy still needs checking.

B The notebook itself recognises image pixels.

Which source or test could settle this?

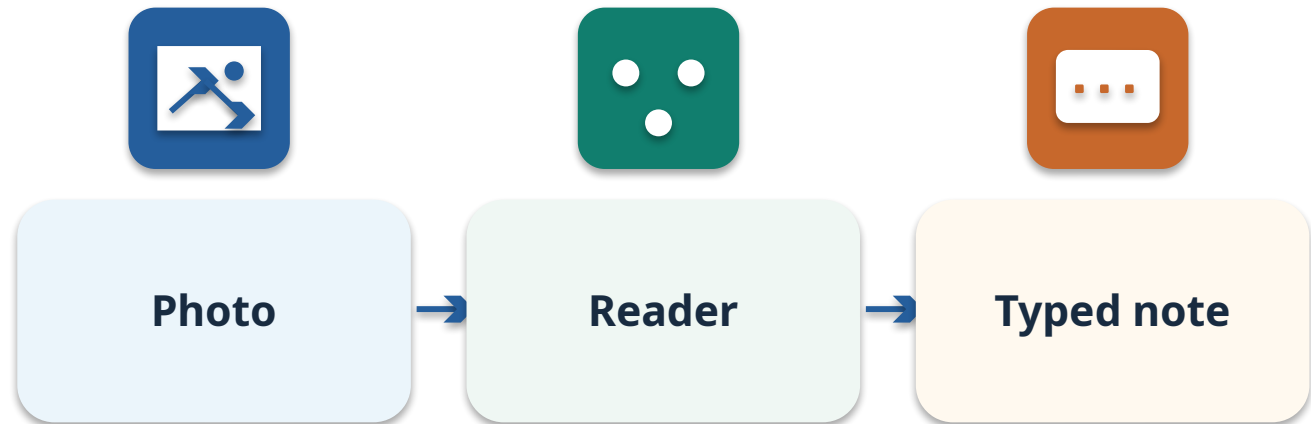
Multimodal input · a familiar picture

ANALOGY / WEEK 10

A person can read a photo and a text note together.

What matches our case, and where does the analogy stop?

A person can read a photo and a text note together.



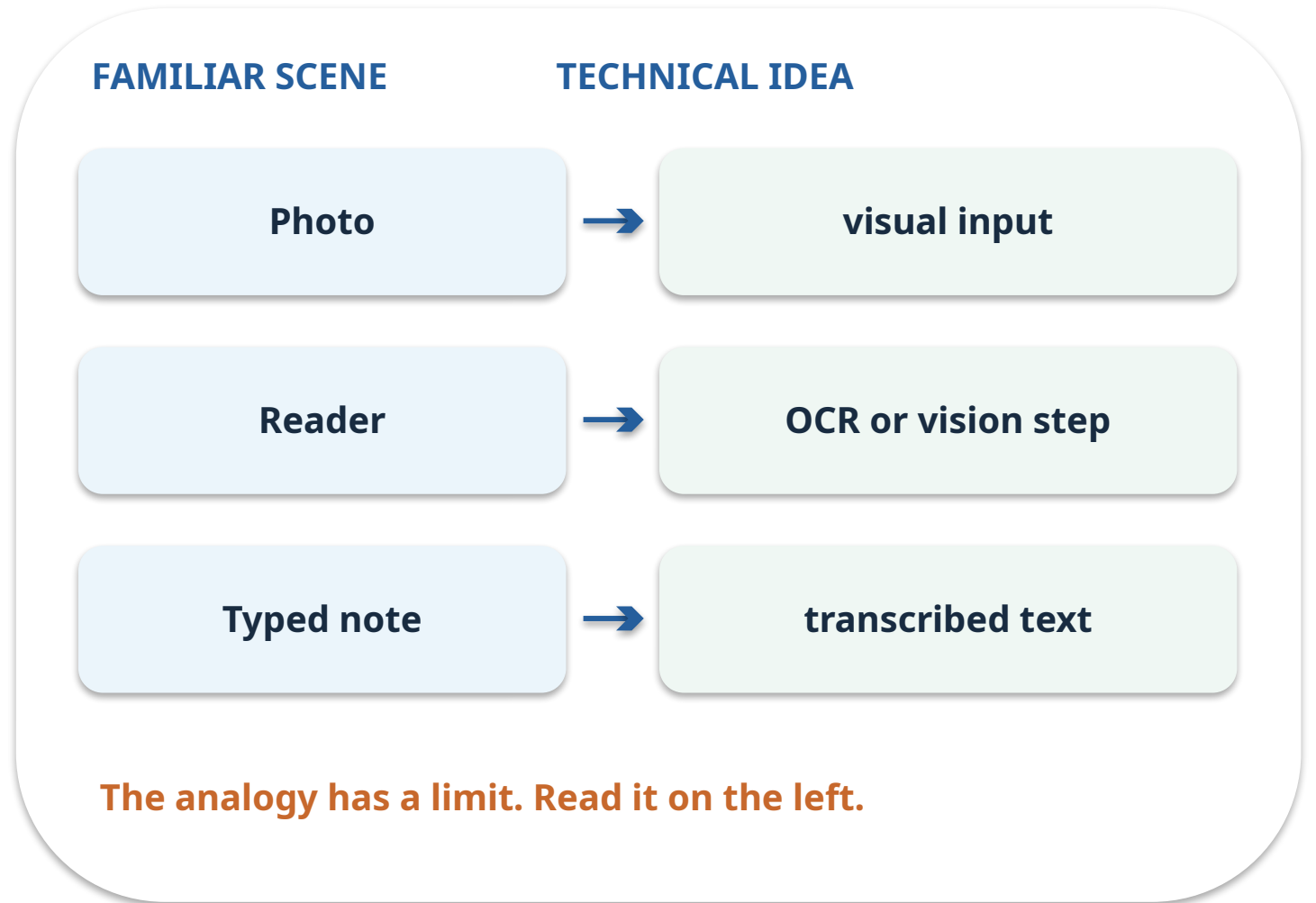
What matches the technical process?

Multimodal input · map the parts

MAPPING / WEEK 10

Match each everyday part to the technical part.

Analogy limit: This notebook uses a prepared transcription; it does not run OCR or test a vision model.



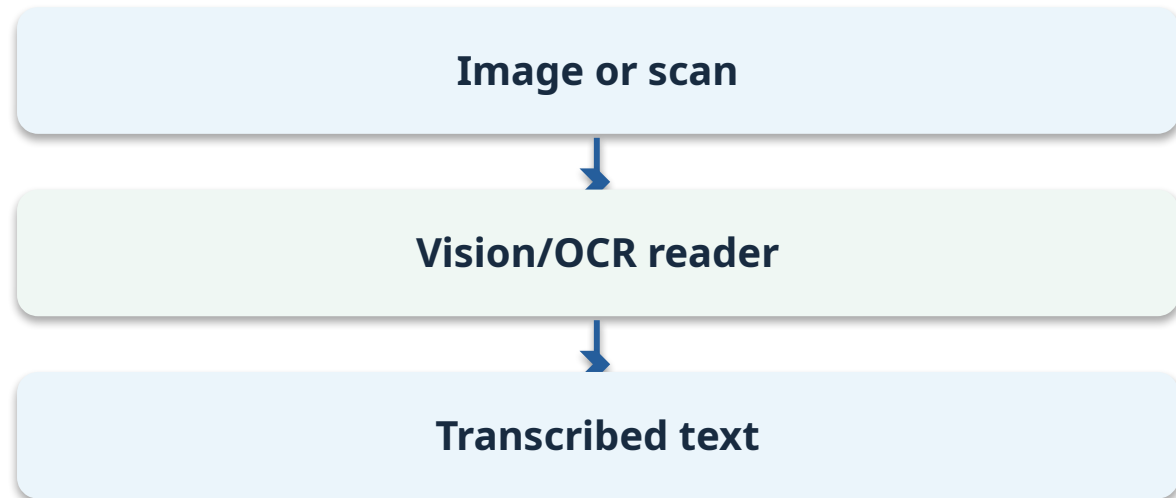
Multimodal input · how it works

MECHANISM / WEEK 10

A system may process text, images or other signals. Here the transcription stands in for an image reader.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Multimodal input · work one case

WORKED / WEEK 10

The prepared text records quantity 2, unit price €20, tax €5 and listed total €54.

Point to the evidence before giving your conclusion.

GIVEN

The prepared text records quantity 2, unit price €20, tax €5 and listed total €54.

QUESTION

What capability is represented by the supplied transcription?

SUPPORTED ANSWER

An upstream reading step, whose accuracy still needs checking.

Multimodal input · answer and explain

REVEAL / WEEK 10

An upstream reading step, whose accuracy still needs checking.

Tempting mistake: The notebook itself recognises image pixels.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The notebook itself recognises image pixels.

SUPPORTED EXPLANATION

An upstream reading step, whose accuracy still needs checking.

This notebook uses a prepared transcription; it does not run OCR or test a vision model.

2. Predict · Extraction

QUESTION / WEEK 10

What does a source pointer let another person do?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Find and verify the value in the original document.

B A correctly formatted field proves its source.

Which source or test could settle this?

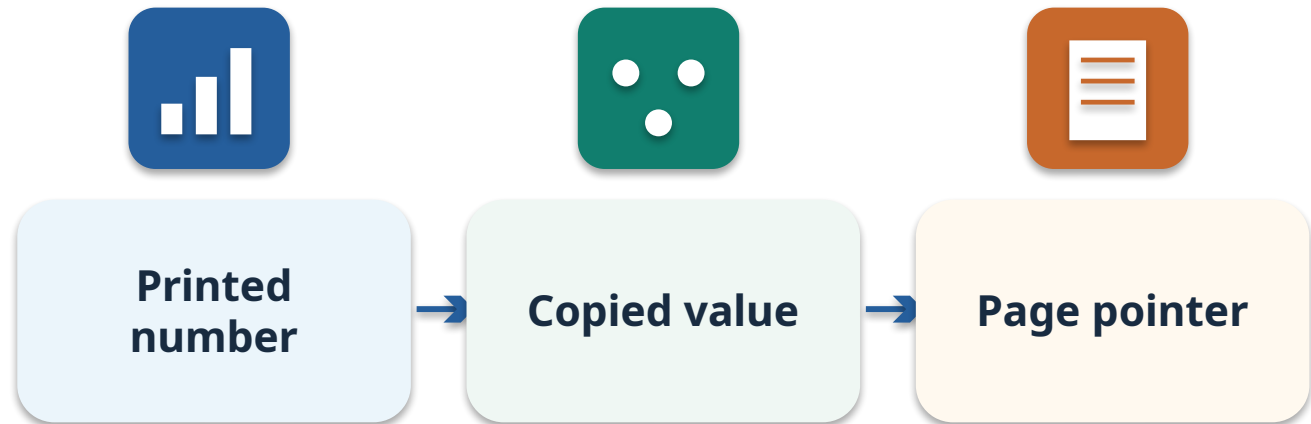
Extraction · a familiar picture

ANALOGY / WEEK 10

An auditor writes the page and line beside every copied number.

What matches our case, and where does the analogy stop?

An auditor writes the page and line beside every copied number.



What matches the technical process?

Extraction · map the parts

MAPPING / WEEK 10

Match each everyday part to the technical part.

Analogy limit: A pointer can be wrong or ambiguous, especially with poor image quality.

FAMILIAR SCENE

Printed number

Copied value

Page pointer

TECHNICAL IDEA

visible field

extraction

source location

The analogy has a limit. Read it on the left.

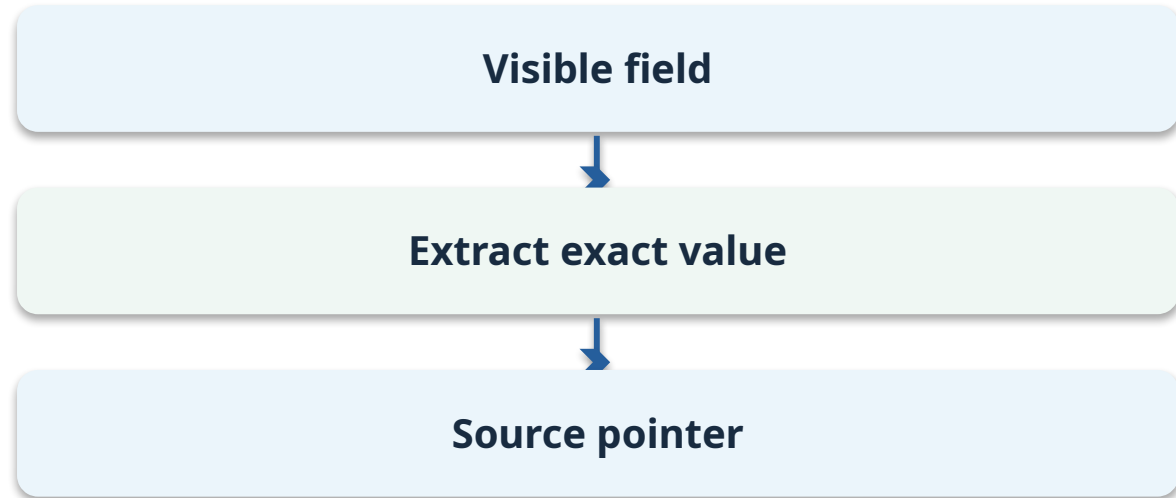
Extraction · how it works

MECHANISM / WEEK 10

Read fields present in the document and retain a pointer to the source.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Extraction · work one case

WORKED / WEEK 10

Record “listed_total = €54” with the invoice field location; keep calculation separate.

Point to the evidence before giving your conclusion.

GIVEN

Record “listed_total = €54” with the invoice field location; keep calculation separate.

QUESTION

What does a source pointer let another person do?

SUPPORTED ANSWER

Find and verify the value in the original document.

Extraction · answer and explain

REVEAL / WEEK 10

Find and verify the value in the original document.

Tempting mistake: A correctly formatted field proves its source.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A correctly formatted field proves its source.

SUPPORTED EXPLANATION

Find and verify the value in the original document.

A pointer can be wrong or ambiguous, especially with poor image quality.

3. Predict · Verification

QUESTION / WEEK 10

Does reading €54 correctly make it a correct payable total?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A No. The independent arithmetic disagrees by €9.

B Accurate OCR implies accurate arithmetic.

Which source or test could settle this?

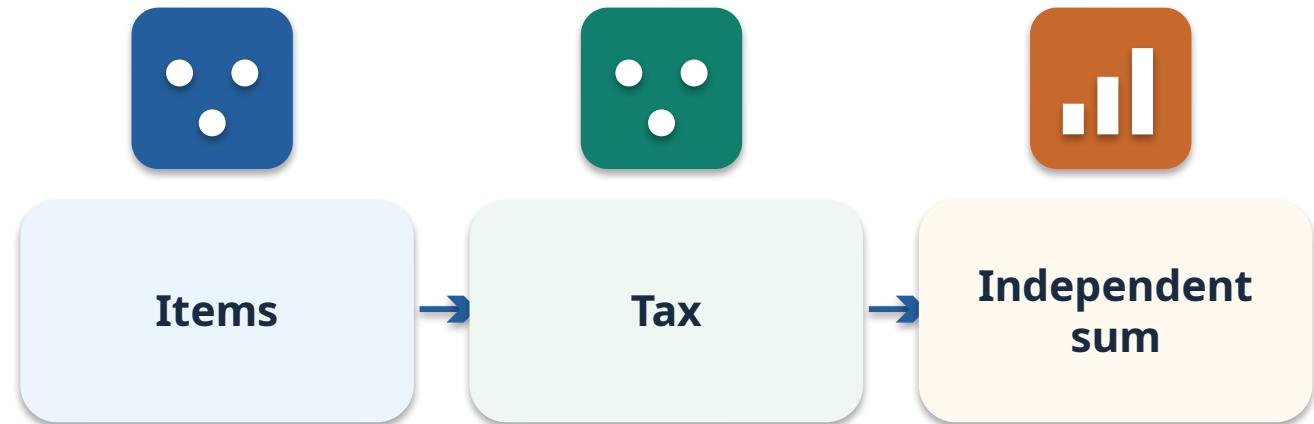
Verification · a familiar picture

ANALOGY / WEEK 10

A cashier recomputes a bill instead of trusting the printed sum.

What matches our case, and where does the analogy stop?

A cashier recomputes a bill instead of trusting the printed sum.



What matches the technical process?

Verification · map the parts

MAPPING / WEEK 10

Match each everyday part to the technical part.

Analogy limit: Real invoices may have discounts and other rules; specify the case assumptions.

FAMILIAR SCENE

Items

Tax

Independent sum

TECHNICAL IDEA

$2 \times \text{€}20$

€5

€45

The analogy has a limit. Read it on the left.

Verification · how it works

MECHANISM / WEEK 10

Recalculate totals and compare against the listed values.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW

Quantity × price



Add tax



Compare listed total

Verification · work one case

WORKED / WEEK 10

$2 \times €20 + €5 = €45$; listed total €54, a €9 discrepancy.

Point to the evidence before giving your conclusion.

GIVEN

$2 \times €20 + €5 = €45$; listed total €54, a €9 discrepancy.

QUESTION

Does reading €54 correctly make it a correct payable total?

SUPPORTED ANSWER

No. The independent arithmetic disagrees by €9.

Verification · answer and explain

REVEAL / WEEK 10

No. The independent arithmetic disagrees by €9.

Tempting mistake: Accurate OCR implies accurate arithmetic.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Accurate OCR implies accurate arithmetic.

SUPPORTED EXPLANATION

No. The independent arithmetic disagrees by €9.

Real invoices may have discounts and other rules; specify the case assumptions.

4. Predict · Uncertainty

QUESTION / WEEK 10

What would be unsafe to store?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A A guessed invoice ID presented as certain.

B A plausible guess should be saved silently.

Which source or test could settle this?

Uncertainty · a familiar picture

ANALOGY / WEEK 10

A reviewer marks an unreadable receipt field with a question mark.

What matches our case, and where does the analogy stop?

A reviewer marks an unreadable receipt field with a question mark.



What matches the technical process?

Uncertainty · map the parts

MAPPING / WEEK 10

Match each everyday part to the technical part.

Analogy limit: Uncertainty needs a clear workflow; marking everything uncertain is unhelpful.

FAMILIAR SCENE

Blurred area

Question mark

Second look

TECHNICAL IDEA

uncertain observation

review flag

verification

The analogy has a limit. Read it on the left.

Uncertainty · how it works

MECHANISM / WEEK 10

Mark unreadable or absent fields for review rather than inventing a value.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW

Blurred or absent field



Mark uncertain



Send for review

Uncertainty · work one case

WORKED / WEEK 10

If INV-08/INV-03 cannot be distinguished, store an uncertain ID and request review.

Point to the evidence before giving your conclusion.

GIVEN

If INV-08/INV-03 cannot be distinguished, store an uncertain ID and request review.

QUESTION

What would be unsafe to store?

SUPPORTED ANSWER

A guessed invoice ID presented as certain.

Uncertainty · answer and explain

REVEAL / WEEK 10

A guessed invoice ID presented as certain.

Tempting mistake: A plausible guess should be saved silently.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A plausible guess should be saved silently.

SUPPORTED EXPLANATION

A guessed invoice ID presented as certain.

Uncertainty needs a clear workflow; marking everything uncertain is unhelpful.

Evidence trail: input → result

WORKED / WEEK 10

Calculated total is €45.

Listed total of €54 triggers review, even if transcription is syntactically valid.

GIVEN

Invoice shows 2 items at €20 plus €5 tax.

QUESTION

Calculated total is €45.

SUPPORTED ANSWER

Listed total of €54 triggers review, even if transcription is syntactically valid.

Notebook: run the prepared example

PRACTICE / WEEK 10

Check two synthetic invoices, including a mismatched total.

Read the original output before editing.

1

Inspect a synthetic source transcription

2

Extract amount, quantity, tax and stated total

3

Calculate the expected total independently

4

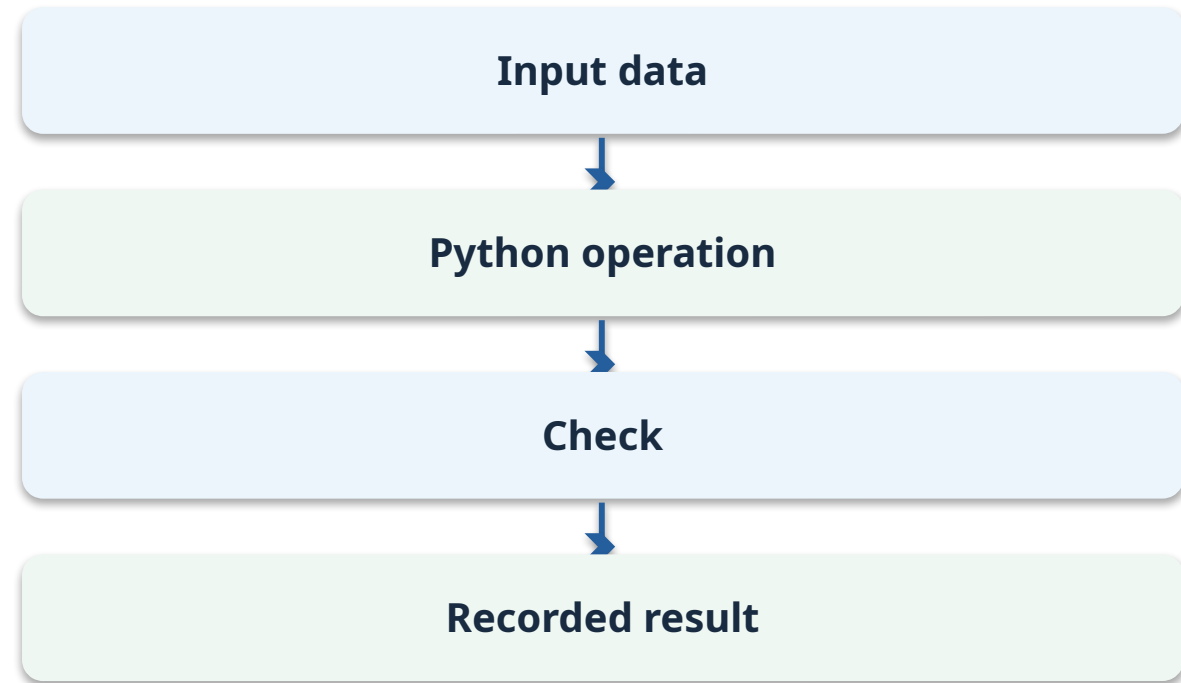
Flag mismatches and retain the source ID

Find the input and check in Python

PRACTICE / WEEK 10

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 10

Correct the listed total and rerun the checker.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 10

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

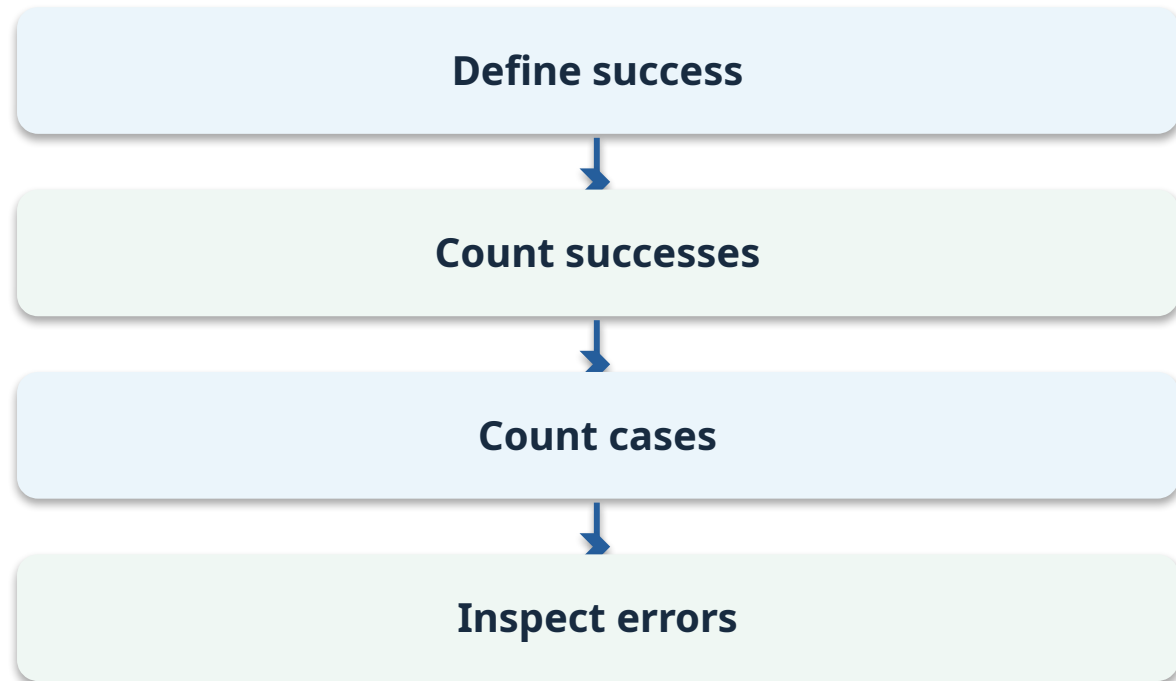
Measure without hiding the denominator

CHECK / WEEK 10

Number of correctly checked fields / fields reviewed

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 10

Plausible wrong claim: The notebook performs OCR on a real image.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The notebook performs OCR on a real image.

SUPPORTED EXPLANATION

This core lab uses prepared transcription to teach verification. Image/OCR integration is an optional extension.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 10

A blurred invoice number looks like INV-08 or INV-03. Which field should be marked uncertain?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

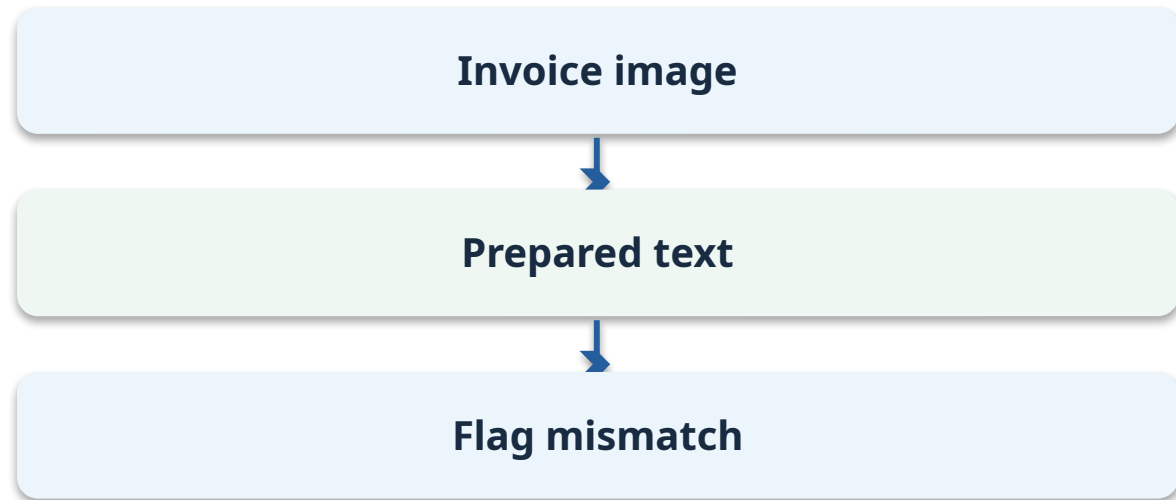
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 10

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 10

What is the difference between extraction and verification?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 10

Calculate the total for 2×20 plus 5 tax.

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

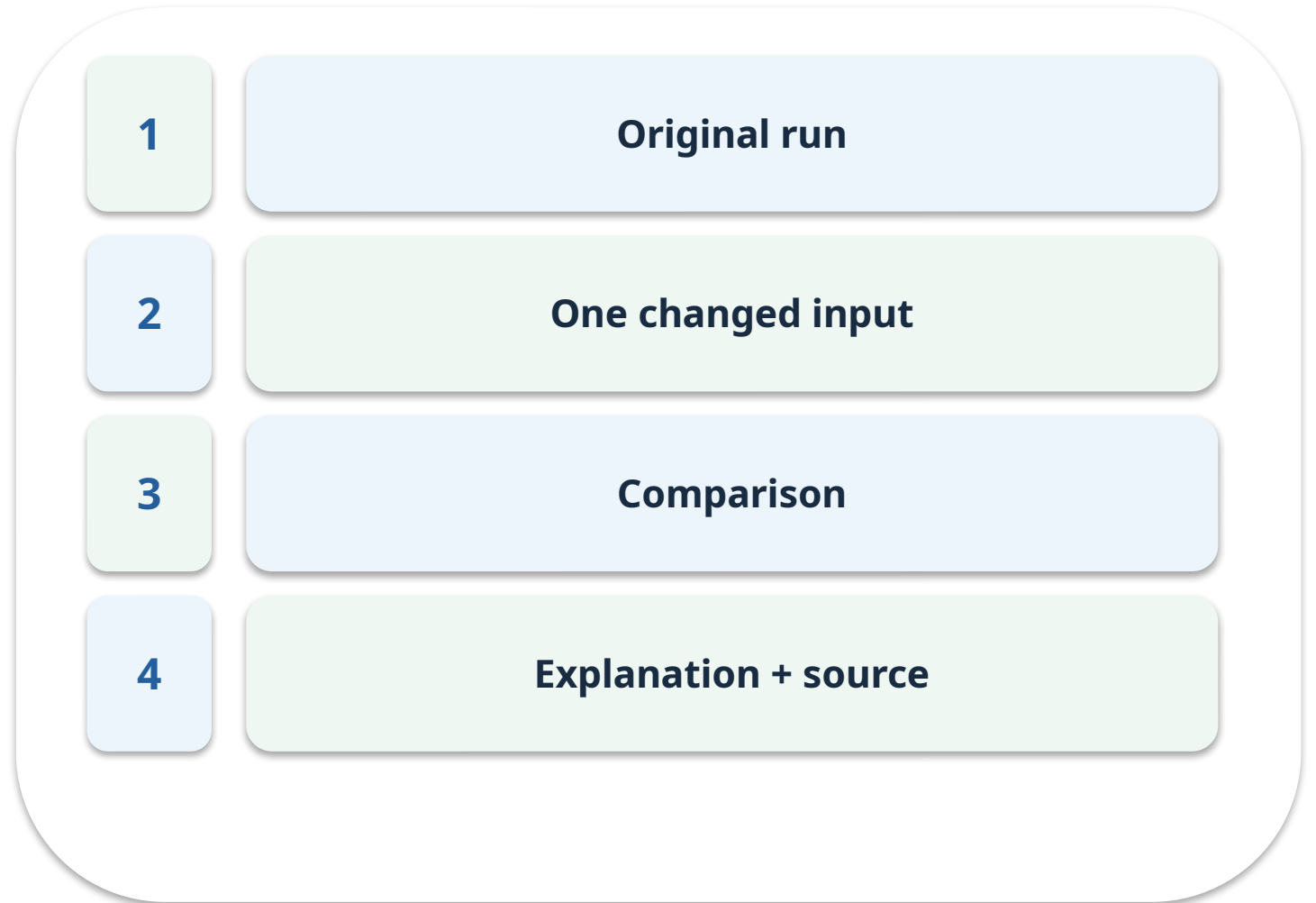
Compare with the Word guide

What to hand in

PRACTICE / WEEK 10

Source-linked field table and calculation.

Include the one changed input and an error explanation.



Three ideas to recall next week

CHECK / WEEK 10

Read the source, preserve field pointers, calculate independently, and flag discrepancies.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

Multimodal input

2

Extraction

3

Verification

4

Uncertainty